



Lecture 8

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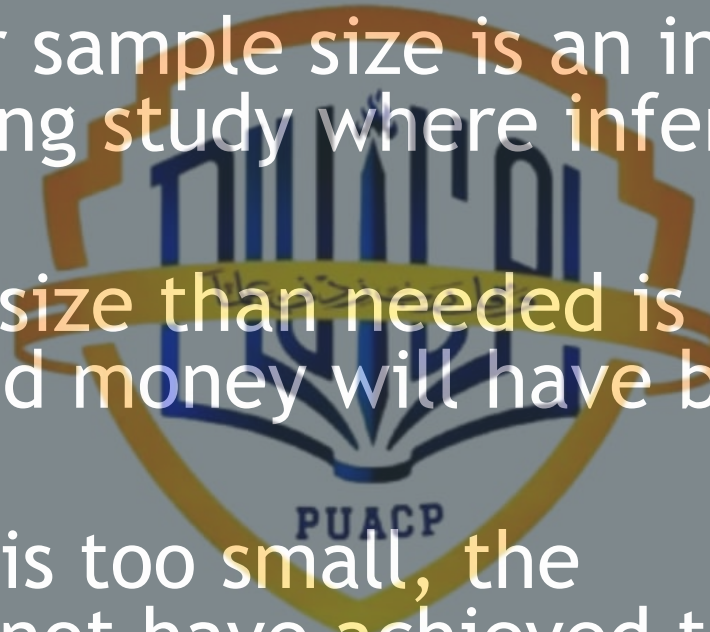
Govt. College, Gujranwala

Choosing an appropriate sample size

- ▶ Collecting data from a sample costs money
- ▶ The larger the sample, the higher the cost.
- ▶ To hold cost down, we want to use as small a sample as possible.
- ▶ But we also want sample to be large enough to provide close estimates of population parameters.

How Large Should The Sample Be?

- ▶ Finding the proper sample size is an integral part of any sampling study where inference need to be made.
- ▶ If a larger sample size than needed is taken, additional time and money will have been spent needlessly.
- ▶ If the sample size is too small, the investigation may not have achieved the desired degree of precision.



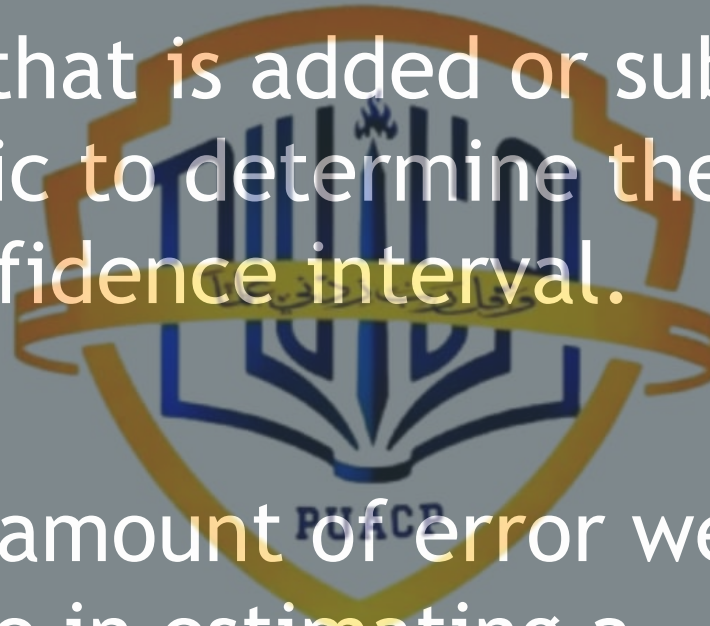
Important Note

The decision of choosing an appropriate sample size depends upon 3 variables.

1. The margin of error the researcher will tolerate.
2. The level of confidence desired.
3. The variation/dispersion of the population being studied.

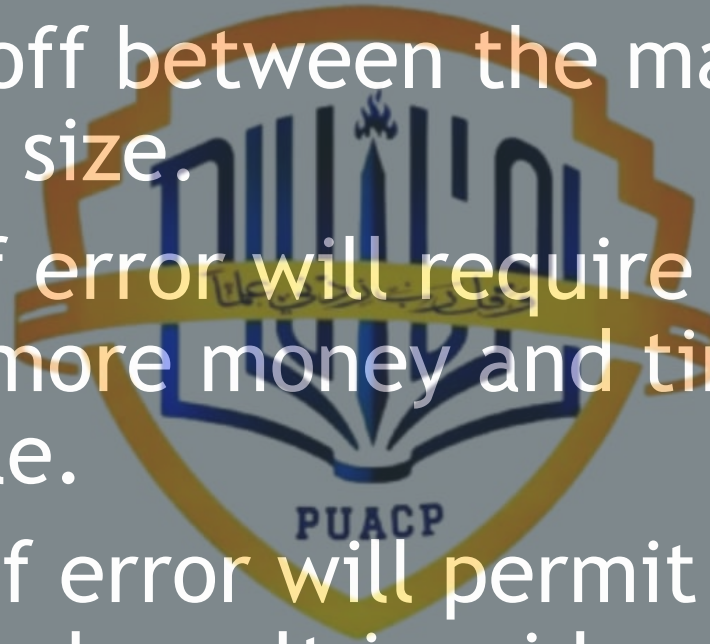
Margin of Error

- ▶ It is the amount that is added or subtracted to sample statistic to determine the end points of the confidence interval.
- ▶ It is actually the amount of error we are willing to tolerate in estimating a population parameter.



Margin of Error

- ▶ There is a trade off between the margin of error and sample size.
- ▶ A small margin of error will require a larger sample size and more money and time to collect the sample.
- ▶ A larger margin of error will permit a smaller sample and result in wider confidence interval.

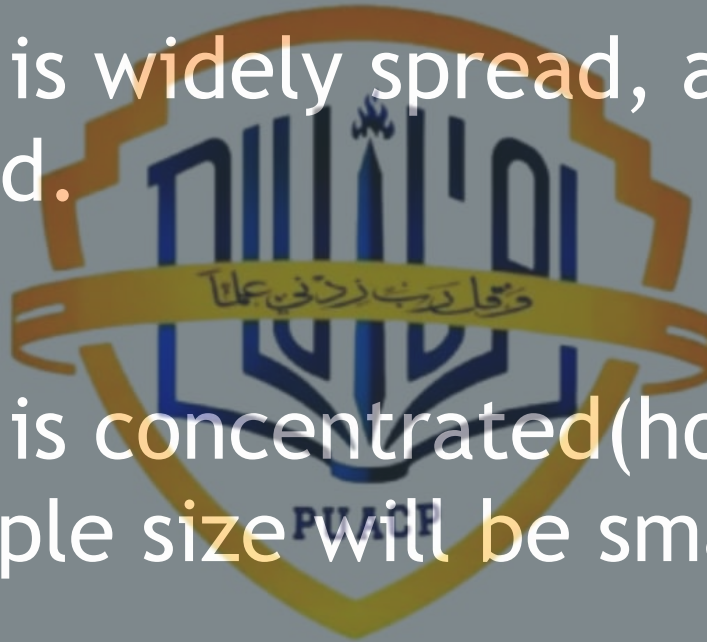


Level of Confidence

- ▶ In working with confidence interval, we logically choose relatively high levels of confidence such as 95% and 99%.
- ▶ To compute sample size, we need the z statistic that corresponds to the chosen level of confidence.
- ▶ The larger sample size correspond with higher levels of confidence.

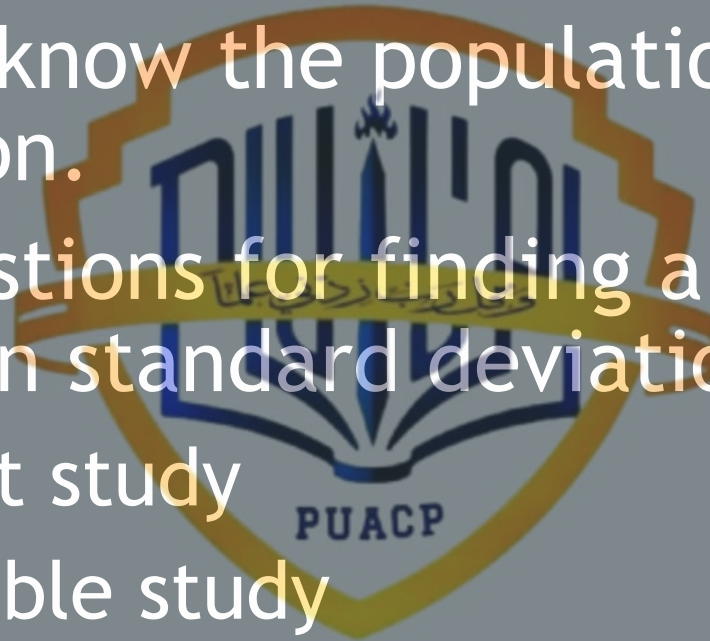
Population Standard deviation

- ▶ If the population is widely spread, a large sample is required.
- ▶ If the population is concentrated (homogeneous) the required sample size will be smaller



Population Standard deviation

- ▶ Often we do not know the population standard deviation.
- ▶ Here are 3 suggestions for finding a value for the population standard deviation.
 1. Conduct a pilot study
 2. Use a comparable study
 3. Use a range base approach.



Conduct a pilot study

- ▶ This is the most common method in which we use a small sample to test the validity of our questionnaire.
- ▶ From this sample we calculate the standard deviation and then use this value as the population standard deviation.

Use a comparable study

- ▶ Use this approach when there is an estimate of the standard deviation from another study.
- ▶ Information from certain state or agency that regularly conduct the required study may provide a reliable value to use for the population standard deviation

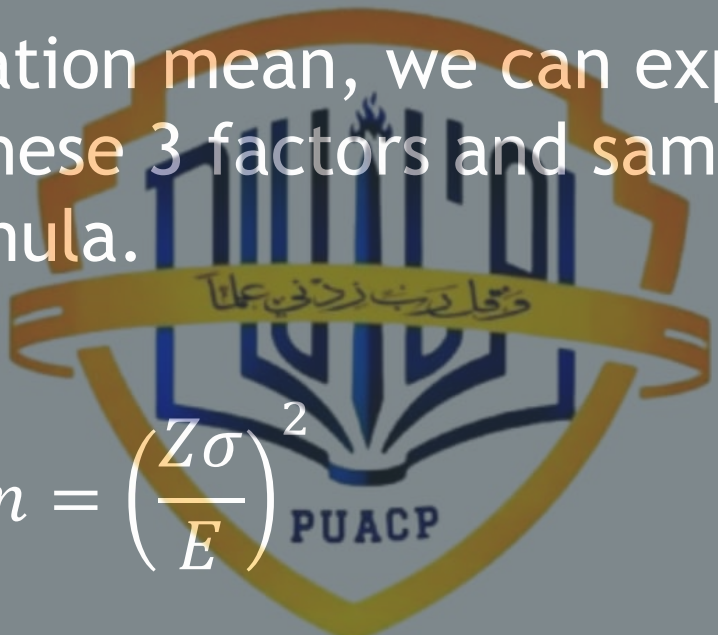


Use a range based approach

- ▶ To use this approach, we need to know an estimate of the largest and smallest values in the population.
- ▶ Empirical rule states that almost all the observations could be expected to be within $(\text{mean}) \pm 3(\text{S.D})$ assuming normal distribution.
- ▶ The distance between largest and smallest observation is 6 standard deviation.
- ▶ We can estimate the standard deviation as one-sixth of the range.

Sample size to estimate a population mean

- To estimate a population mean, we can express the interaction among these 3 factors and sample size in the following formula.

The logo of the Punjab University of Agriculture, Chaudhary Charan Singh (PUACP) is centered in the background. It features a shield with a book and a torch, with the university's name in Urdu and English. The acronym 'PUACP' is at the bottom.
$$n = \left(\frac{Z\sigma}{E} \right)^2$$

Sample size to estimate a population Proportion

- ▶ To estimate a population proportion, same 3 factors need to be specified in the following manner.

$$n = \pi(1 - \pi) \left(\frac{Z}{E} \right)^2$$

- ▶ The choices for z statistic and margin of error are the same but population variance is represented by $\pi(1-\pi)$.
- ▶ If a reliable value of π can not be found then a value of 0.5 should be used for π .



Thanks